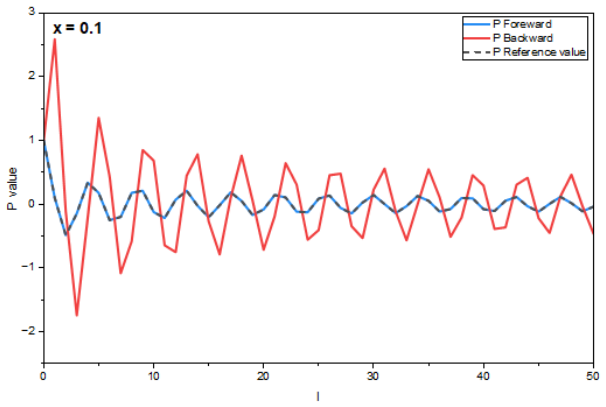
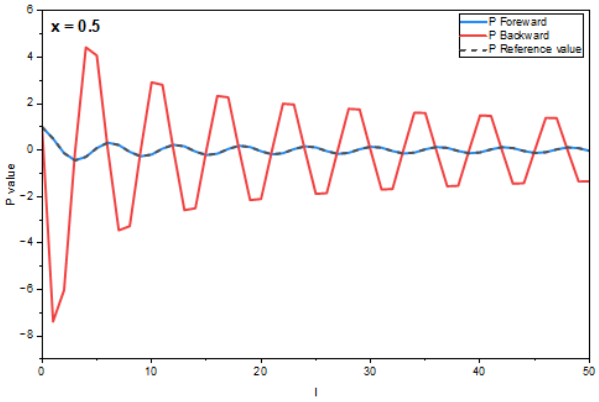
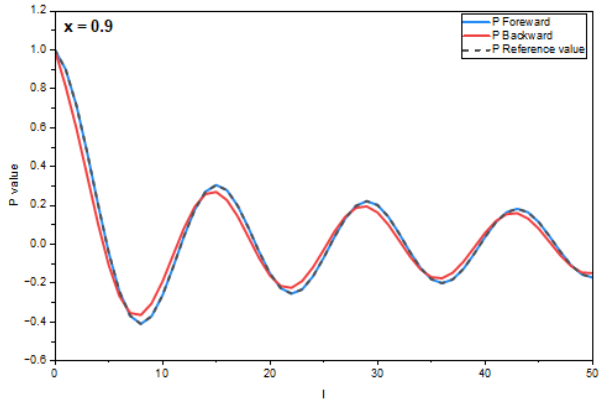
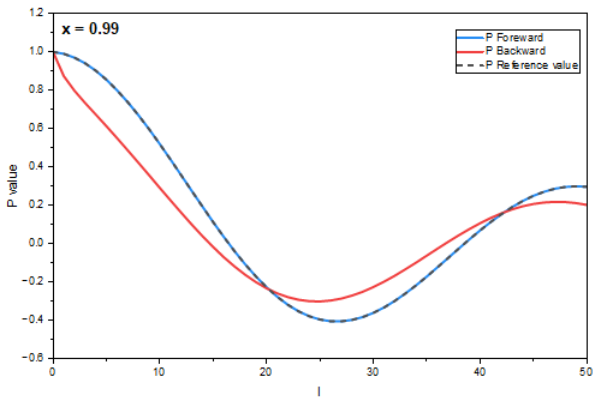
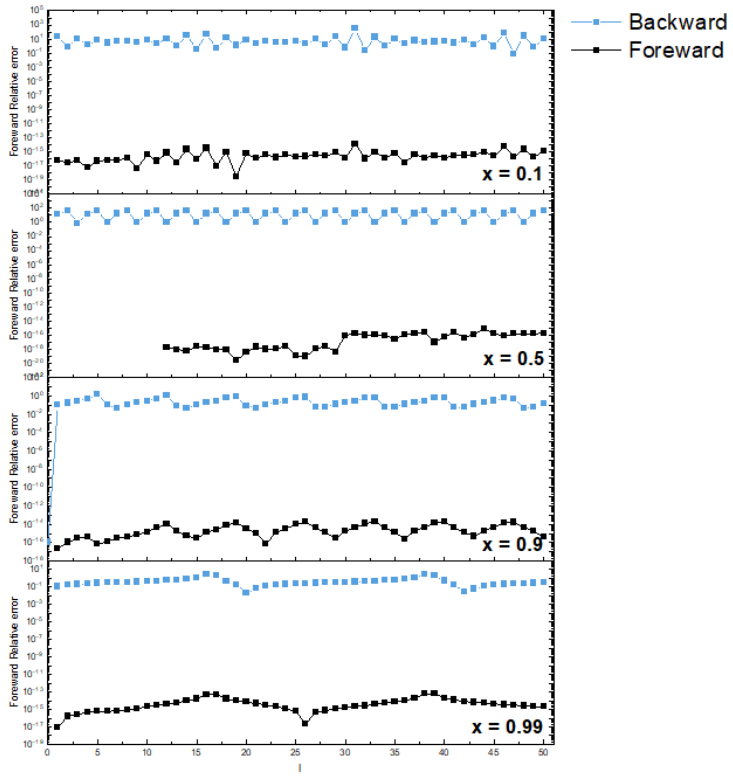
Homework 5











The results agree with the expected behaviour described in the assignment. The values calculated using the forward recurrence closely follow the high-precision reference values for all the tested values of x. On the other hand, the backward recurrence does not generally reproduce the reference Legendre polynomials.
This result is also confirmed by the relative errors. The forward recurrence shows very small errors, while the errors obtained with the backward recurrence are several orders of magnitude larger.
Therefore, for the tested values of x and for $l<50$, the forward recurrence appears numerically stable, whereas the arbitrary backward recurrence does not reliably recover $P_l(x)$.